Integrating R and LaTeX in a single source document with the knitr package: An example of reporting reproducible research

(version 4.4)

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Abstract

Here you can put your abstract...

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1 Introduction

This report is an example of reproducible research using the function `knit()`, in the R package knitr, and LaTeX. For an introduction to knitr, I recommend the website of the package [1].

1.1 Toy example

The aim of this document is to provide a minimal example of a `Rnw` source file that integrates code of both R and LaTeX, which result is this PDF file. The statistical analysis shown in this document, a linear regression fitting, is only for illustration purposes on how `knitr` works and should not be considered for statistical purposes. In Section 2 we explore data graphically. Then, in Section 3, we fit and interpret a linear regression model. The R code used is shown in Appendix A. Appendix A.1 shows the R packages we need to get the PDF for this document.

2 Data

Figure 1[1] shows a graphical representation of the relationship between the wind speed and ozone levels, where we can see a possible linear association:

$$\text{Ozone} = \beta_0 + \beta_1 \text{Wind speed} \tag{1}$$

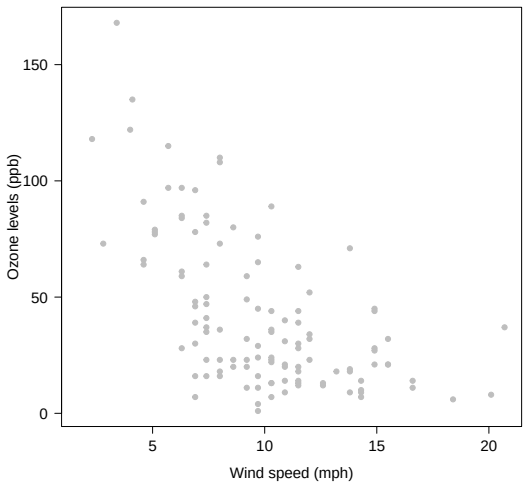


Figure 1: Scatter plot of ozone levels vs. wind speed.

3 Modeling

```
##             Estimate Std. Error  t value     Pr(>|t|)
## (Intercept) 96.872895  7.2386887 13.38266 3.992068e-25
## Wind        -5.550923  0.6904021 -8.04013 9.271974e-13
```

In Section 2 we have seen that a there is a possible linear relationship between the wind speed and the ozone levels, under model (1). Thus, we can try to fit a linear regression model[2]. Specifically, Table 1 shows the estimated coefficients of the linear model[3]. Under this model, a one unit increase in the wind speed is associated to a decrease of 5.55 units in the mean of ozone levels. Figure 2 shows the data as well as the fitted model.

[1] See Appendix A.2.
[2] See R code in Appendix A.3.
[3] The vignette "The xtable Gallery" [2] provides a lot of examples for table customization.

	Estimate	Std. Error	t-value	p-value
(Intercept)	96.87	7.24	13.38	0.0000
Wind	-5.55	0.69	-8.04	0.0000

Table 1: Linear regression model coefficients for wind and ozone association.

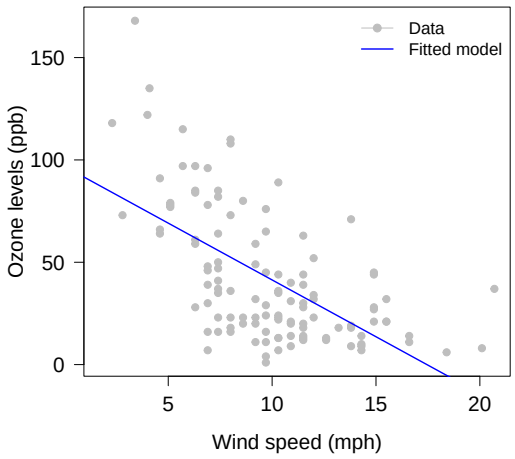


Figure 2: Scatter plot of ozone levels vs. wind speed and the fitted linear regression model for the mean of ozone levels as a function of wind speed. In the fitted model, $\hat{\beta}_0 = 96.87$ ppb and $\hat{\beta}_1 = -5.55$ ppb/mph.

4 Extracting R code

You can extract the R code included in the `Rnw` source document to a R file by using the `purl()` function in the knitr package:

```
> library(knitr)
> purl("knitr_example.Rnw")
```

Read `?purl` to see how the R code extraction can be customized (e.g., which chunks must be extracted, split R files, file name or file documentation)

Bibliography

[1] Yihui Xie. knitr: A General-Purpose Package for Dynamic Report Generation in R, 2024. URL https://yihui.org/knitr/. R package version website.

[2] D. Dahl, D. Scott, C. Roosen, A. Magnusson, and J. Swinton. The xtable Gallery, 2019. URL https://cran.r-project.org/web/packages/xtable/vignettes/xtableGallery.pdf. R package vignette.

A R code

A.1 Libraries related to knitr

```
> ### libraries related to knit:
> library(knitr)
> library(xtable)     # to create nice LaTeX tables
```

A.2 Plotting data

```
> ### scatter plot ozone vs. wind:
> par(las = 1, cex.lab = 1.3, cex.axis = 1.3)
> plot(airquality$Wind,
+      airquality$Ozone,
+      xlab = "Wind speed (mph)",
+      ylab = "Ozone levels (ppb)",
+      pch = 19,
+      col = "gray")
```

A.3 Fitting the linear regression model

```
> ### linear regression model fitting:
> mod <- lm(Ozone ~ Wind, data = airquality)
> ### get coefficients table:
> coefs <- summary(mod)$coefficients
> show(coefs)
> ### get beta of interest:
> beta <- coefs["Wind", "Estimate"]
```

A.4 Plotting the fitted model

```
> # scatter plot ozone vs. wind again (recycling code in chunk dataplot):
>
> # we increase the font sizes to compensate the reduction that dev='tikz'
> # introduces:
> par(cex = 1.4)
>
> # reuse the previous chunk named "dataplot":
>
> ### scatter plot ozone vs. wind:
> par(las = 1, cex.lab = 1.3, cex.axis = 1.3)
> plot(airquality$Wind,
+      airquality$Ozone,
+      xlab = "Wind speed (mph)",
+      ylab = "Ozone levels (ppb)",
+      pch = 19,
+      col = "gray")
>
> # add the fitted model line:
> abline(mod, col = "blue", lwd = 2)
>
> # add legend:
> legend("topright",
+        legend = c("Data", "Fitted model"),
+        col = c("gray", "blue"),
+        lwd = c(0, 2),    # line width
+        pch = c(19, NA), # point symbol
+        bty = "n",        # no box around the legend
+        cex = 1.1)        # font size (multiplier)
```

A.5 Exporting the table to LaTeX

```
> # exporting coefficients table to LaTeX format:
> modtable <- xtable(coefs,
+                    digits = c(0, 2, 2, 2, 4),
+                    caption = "Linear regression model coefficients for wind and ozone association.",
+                    label = "tab:coeftable")
> # cosmetics for table head (LaTeX slash (\) must be doubled inside a R chunk):
> colnames(modtable) <- c("\\multicolumn{1}{c}{\\bf Estimate}",
+                         "\\multicolumn{1}{c}{\\bf Std.~Error}",
+                         "\\multicolumn{1}{c}{\\(\\boldsymbol{t}\\)\\bf{-value}}",
+                         "\\multicolumn{1}{c}{\\(\\boldsymbol{p}\\)\\bf{-value}}")
>
> # format and print the table:
> print(modtable,
+       table.placement = "ht",
+       size = "footnotesize",
+       include.rownames = TRUE,
+       include.colnames = TRUE,
+       floating = TRUE,                  # whether \begin{table} should be created (TRUE) or not (FALSE)
+       sanitize.text.function = force,   # important to treat content of columns as latex function
+       booktabs = TRUE)                  # requires \usepackage{booktabs} in the preamble of the document
```